

EcoSyz

Door – To – Door Garbage Collection Smart Reporting System



Project Team

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ABBREVIATIONS USED

Abbreviation	Meaning
UML	Unified Modeling Language
SMS	Short Message Service
SaaS	Software as a Service
REST	Representational State Transfer
RBAC	Role-Based Access Control
QR	Quick Response Code
PDF	Portable Document Format
OTP	One-Time Password
NLP	Natural Language Processing
MERN	MongoDB, Express, React, Node.js
JWT	JSON Web Token
GPS	Global Positioning System
E2E	End-to-End
CSS	Cascading Style Sheets
API	Application Programming Interface

ABSTRACT

Keywords: EcoSyz, Garbage Collection, Smart Waste Management, QR Code, GPS Geofencing, Voice Assistance, MongoDB, React Native

EcoSyz is a digital smart reporting system designed to improve the monitoring, verification, and management of door-to-door garbage collection operations. The system connects citizens, sanitation workers, Panchayat administrators, and Super Admin users through web and mobile interfaces. It supports voice-assisted collection logging, QR-based verification, GPS geofencing, digital attendance, citizen grievances, pickup requests, dashboards, analytics, and PDF/Excel reporting. A tenant-aware cloud backend provides centralized data management, while offline synchronization supports field operation in areas with unstable connectivity. The implemented workflows demonstrate how web, mobile, GPS, QR, voice input, and cloud services can be combined into a practical waste-management reporting platform.

CHAPTER 1: SYSTEM ANALYSIS

1.1 INTRODUCTION

System analysis identifies the operational requirements of a system and converts them into clear software functions. For EcoSyz, the analysis focuses on the movement of information between households, field workers, Panchayat administrators, and supervisory users. The system must support daily collection work while producing reliable records for monitoring and reporting.

1.1.1 PROJECT BACKGROUND

Door-to-door garbage collection involves repeated field activities such as route assignment, household or dustbin identification, attendance, waste measurement, collection logging, and complaint handling. Manual registers make these activities difficult to monitor and make later reporting dependent on individual records.

EcoSyz addresses this operational gap through a connected digital workflow. A worker can authenticate, verify duty location, collect waste, record the collection using voice or QR-supported workflows, and synchronize the information with the backend. Administrators can review the information through dashboards and reports, while citizens can access schedules and submit grievances.

1.2 AIM AND OBJECTIVES

The aim of EcoSyz is to develop a digital smart reporting system that improves the monitoring, verification, and management of door-to-door garbage collection operations.

1. Provide voice-assisted field logging so workers can enter collection information without extensive typing.
2. Use QR codes and GPS validation to provide proof of collection and location-based verification.
3. Provide digital attendance management with geofencing and automated absent reconciliation.
4. Allow citizens to view collection information and submit grievances or special pickup requests.
5. Provide administrators with dashboards, data management, analytics, and PDF/Excel reporting.
6. Maintain tenant-separated data through a common cloud backend.

1.3 RESEARCH QUESTION

How can a multi-platform digital ecosystem improve the operational efficiency, field accountability, and citizen participation of waste-management services within Village Panchayats?

1.4 SYSTEM REQUIREMENTS

Table 1.1. System requirements

Requirement Area	Requirement
User roles	Citizen, Sanitation Worker, Panchayat Admin, Super Admin
Web applications	Citizen portal, Panchayat Admin portal, Super Admin portal
Mobile application	Labour application for attendance and collection
Backend	REST API with authentication and background jobs
Database	MongoDB Atlas with tenant-aware records
Field validation	GPS geofence and QR verification
Data entry	Voice-assisted and form-based collection logging
Reporting	PDF and Excel export
Reliability	Offline caching and synchronization

CHAPTER 2: EXISTING SYSTEM AND PROPOSED SYSTEM

2.1 EXISTING SYSTEM

The existing process commonly depends on manual attendance records, handwritten collection information, telephone-based complaints, and separate records maintained by different operational teams. These methods can make it difficult to obtain a single current view of collection performance.

2.2 LIMITATIONS OF EXISTING SYSTEM

Table 2.1. Limitations of existing system

Limitation	Effect
Manual records	Data entry and later consolidation require additional effort.
Limited real-time visibility	Supervisors cannot easily confirm current route and collection status.
Unverified field activity	Paper records do not provide location-based proof of service.
Typing during collection	Field workers may find screen typing inconvenient during active collection work.
Complaint fragmentation	Phone or paper complaints are difficult to track from submission to resolution.
Separate systems	Information can remain divided between different operational records.

2.3 COMPARATIVE FEATURE MATRIX

Table 2.2. Comparative feature matrix

Feature	Manual Process	EcoSyz
Collection logging	Paper/manual entry	Voice-assisted and digital logging
Proof of service	No location proof	QR + GPS validation
Attendance	Register	Geofenced mobile attendance
Complaint handling	Phone/paper	Tracked digital grievance
Reporting	Manual compilation	PDF/Excel generation
Data storage	Separate records	Central cloud database
Offline operation	Not supported	Local caching and synchronization

CHAPTER 3: SYSTEM DESIGN AND IMPLEMENTATION

3.1 SYSTEM DESIGN

EcoSyz is organized into presentation, application, and data layers. Web and mobile interfaces communicate with the backend API. The backend performs authentication, validation, business operations, background scheduling, and database access. This separation allows each user role to interact with only the functions required for its work.

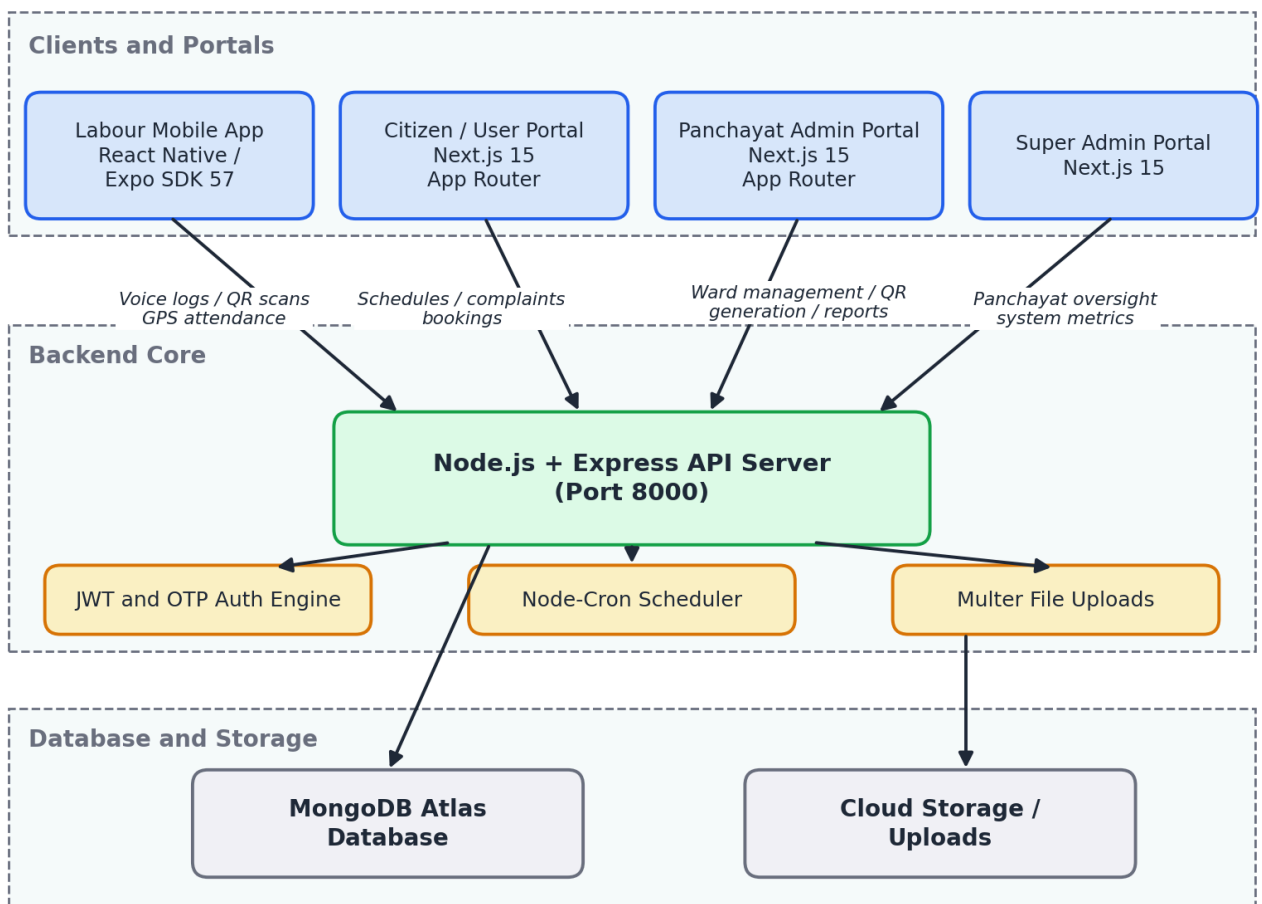


Figure 3.1. EcoSyz System Architecture

3.2 SYSTEM USE CASE DIAGRAM

EcoSyz Smart Waste Platform — System Use Case Diagram

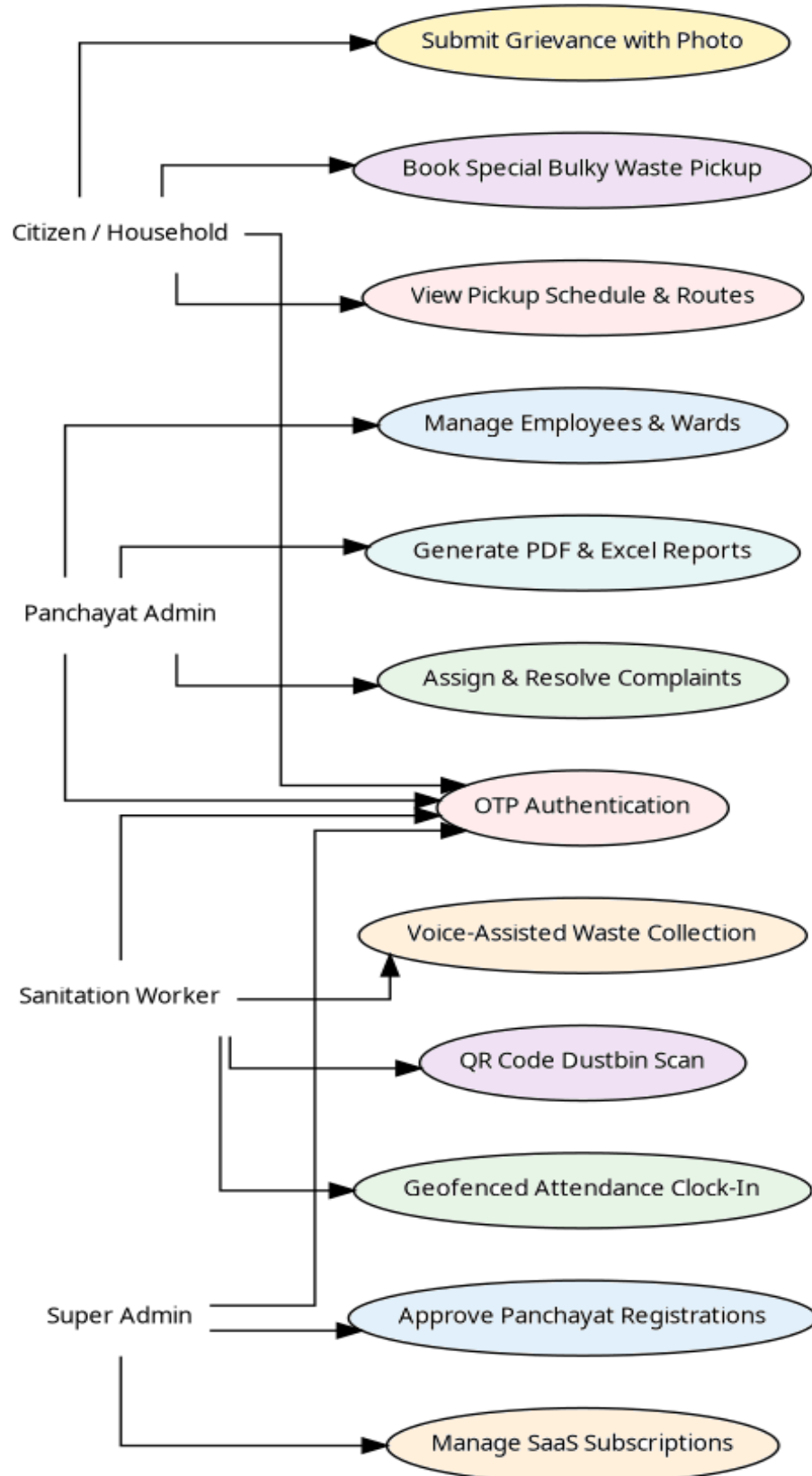


Figure 3.2. System Use Case Diagram

3.3 VOICE-ASSISTED COLLECTION SEQUENCE

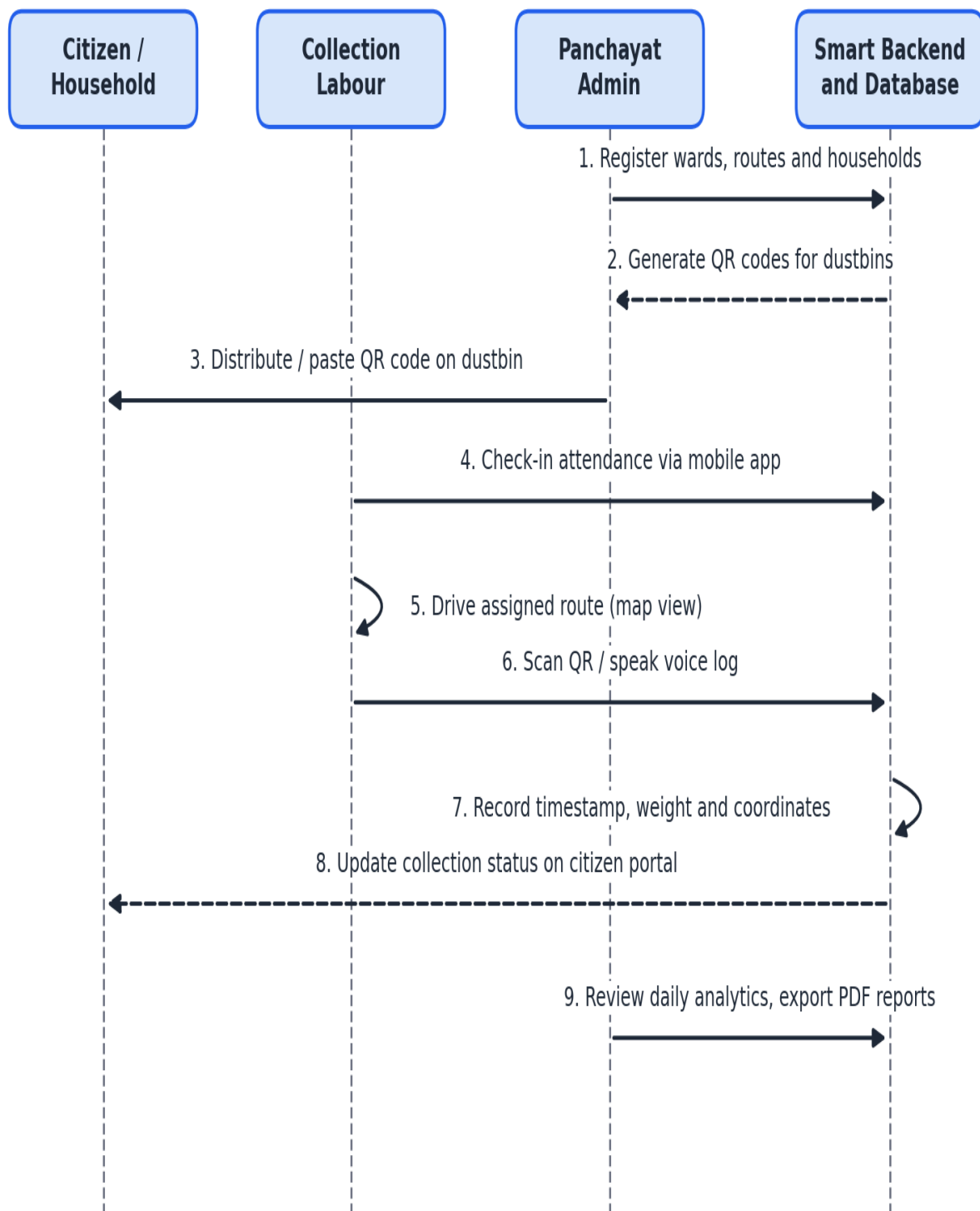


Figure 3.3. Sequence Diagram for Voice-Assisted Waste Collection

3.4 ATTENDANCE AND MIDNIGHT CRON WORKFLOW

Geofenced Attendance and Midnight Cron Workflow

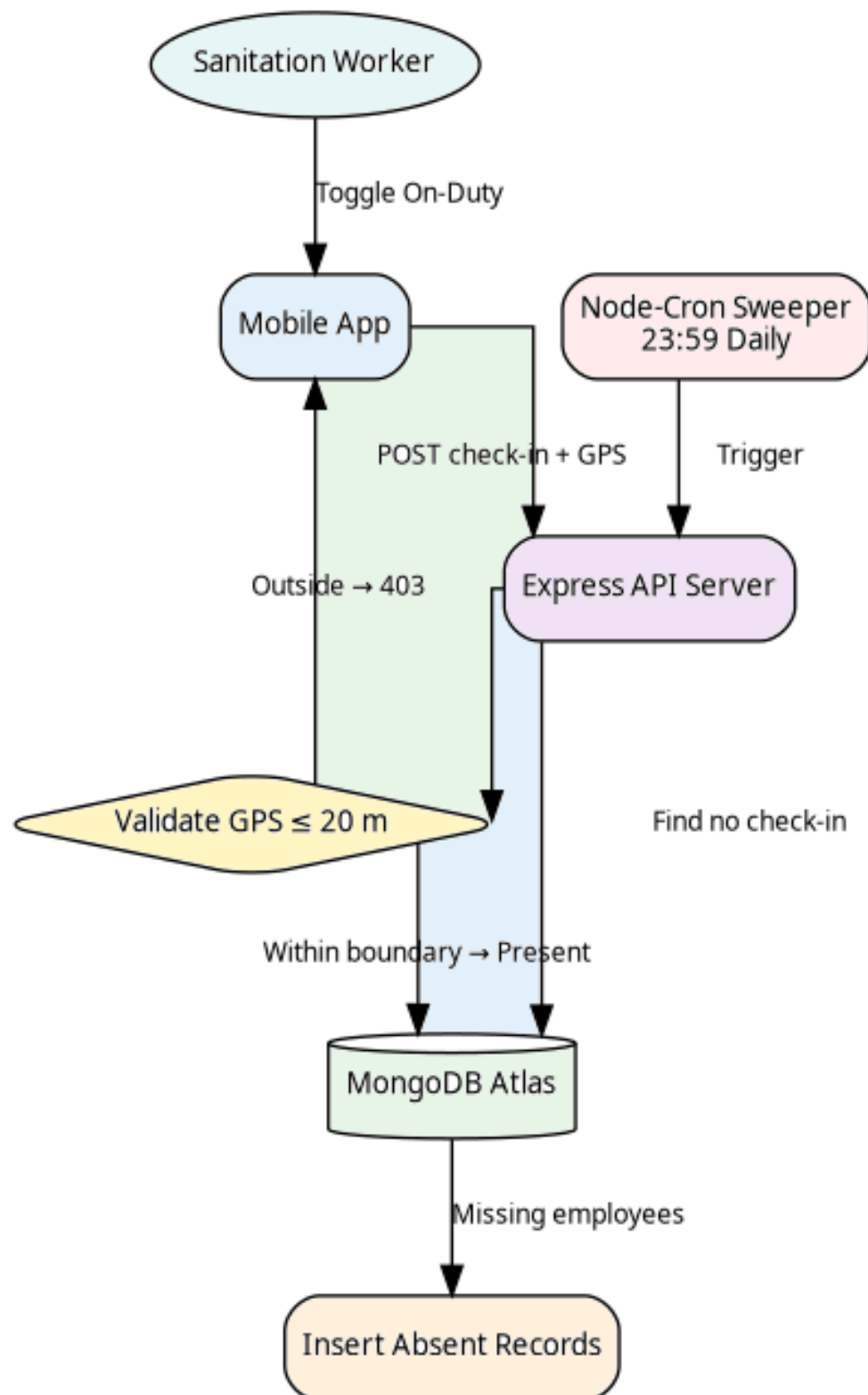


Figure 3.4. Geofenced Attendance and Midnight Cron Workflow

3.5 FIELD WORKER ACTIVITY DIAGRAM

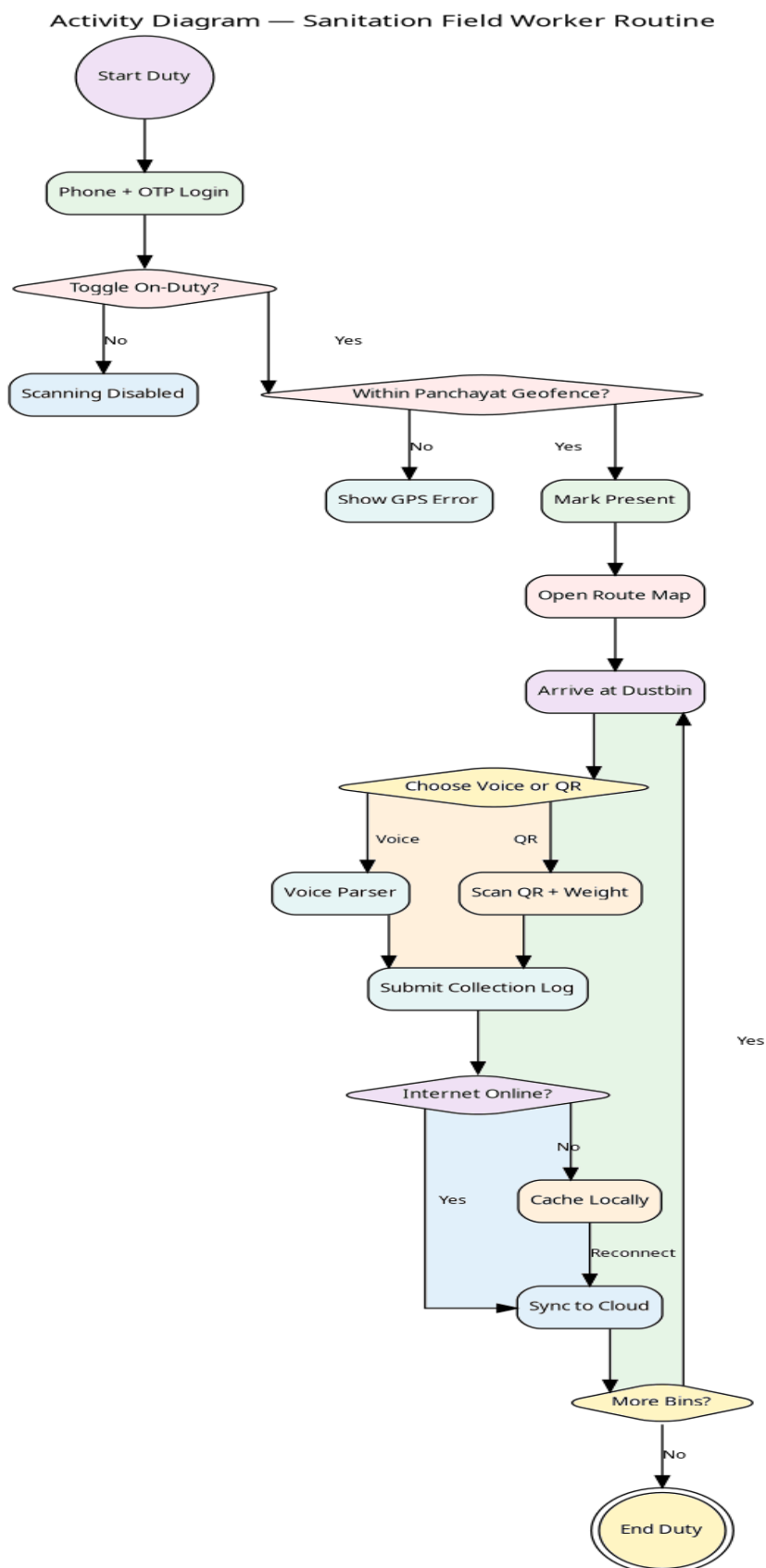


Figure 3.5. Activity Diagram for Sanitation Field Worker

3.6 ADMINISTRATIVE ACTIVITY DIAGRAM

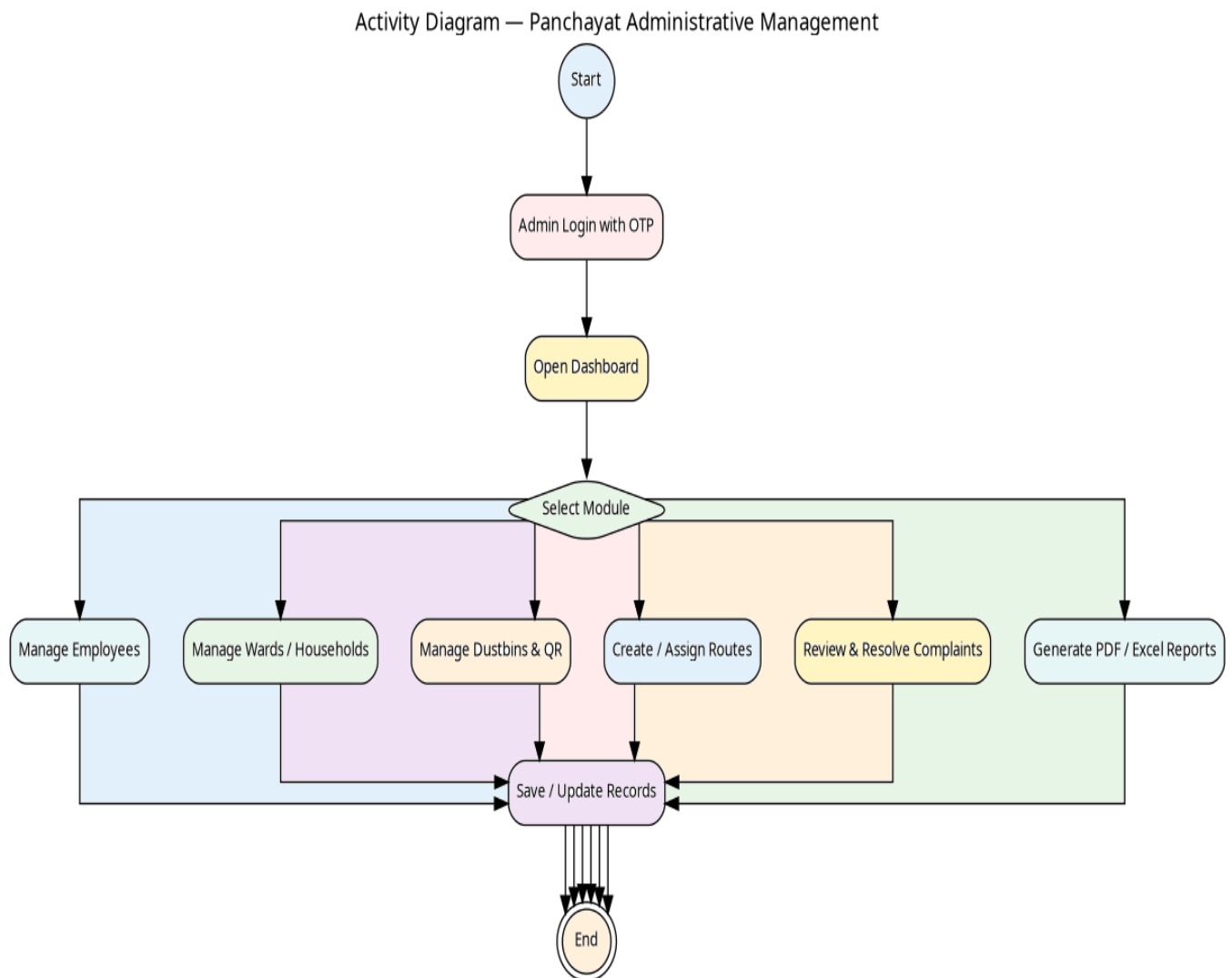


Figure 3.6. Activity Diagram for Administrative Management

3.7 CITIZEN ACTIVITY DIAGRAM

Activity Diagram — Citizen Grievance and Pickup Booking

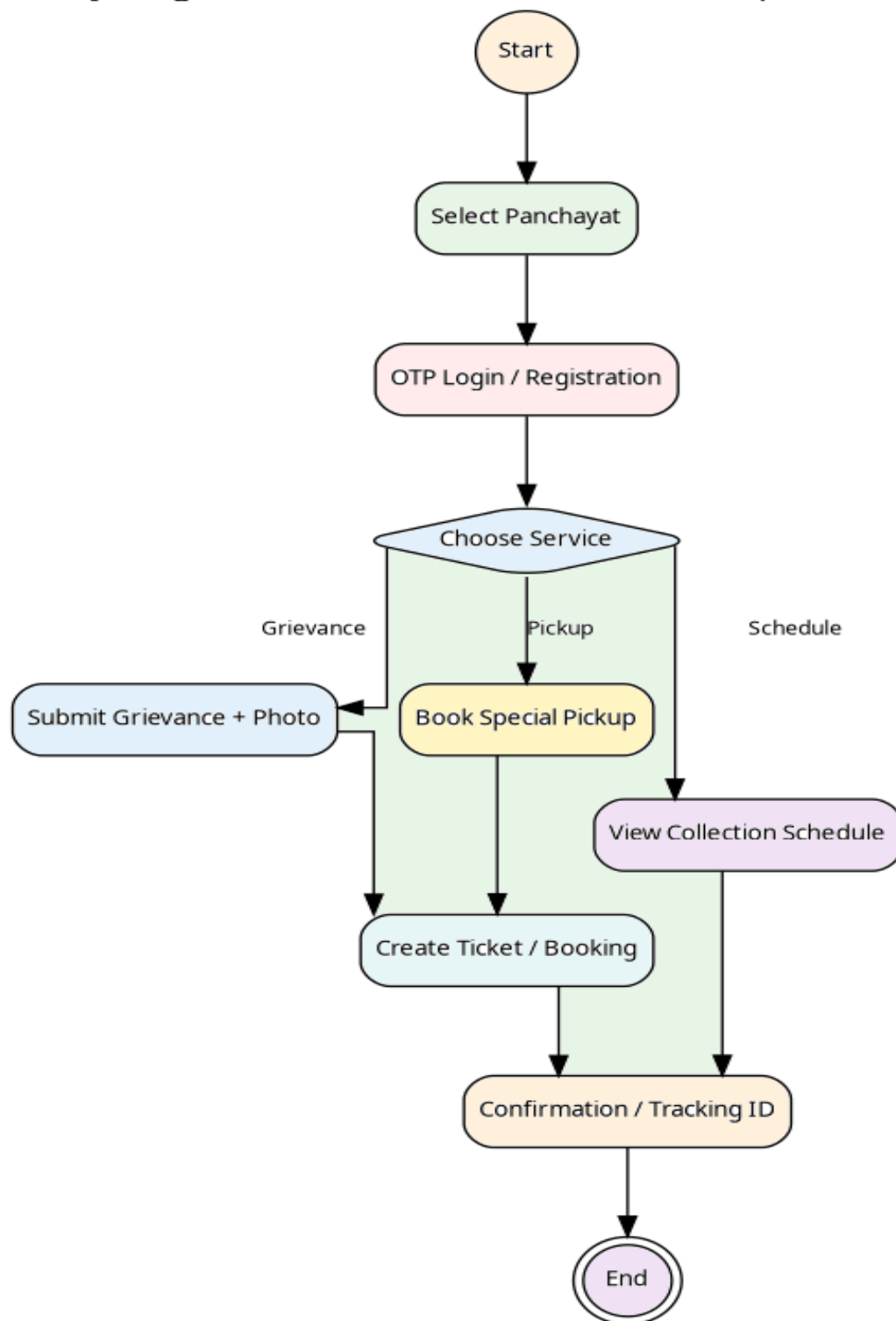


Figure 3.7. Activity Diagram for Citizen Grievance and Pickup Booking

3.8 DATABASE ER DIAGRAM

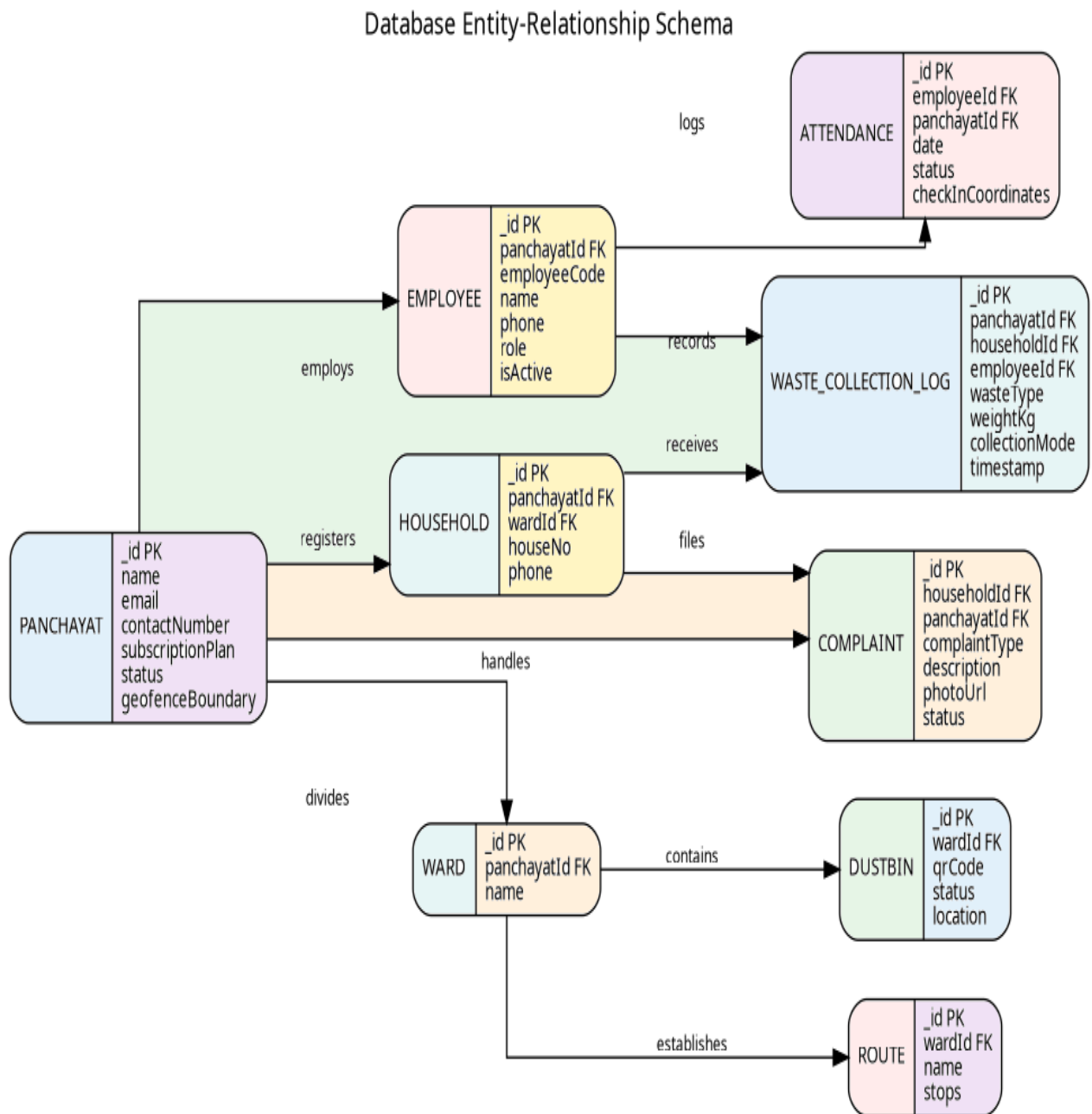


Figure 3.8. Database Entity-Relationship Schema

3.9 DATA FLOW DIAGRAMS (DFD)

The Data Flow Diagrams describe how information moves between external stakeholders, the EcoSyz platform, operational processes, and persistent data stores. Level 0 defines the system boundary, while Level 1 decomposes the major operational flows.

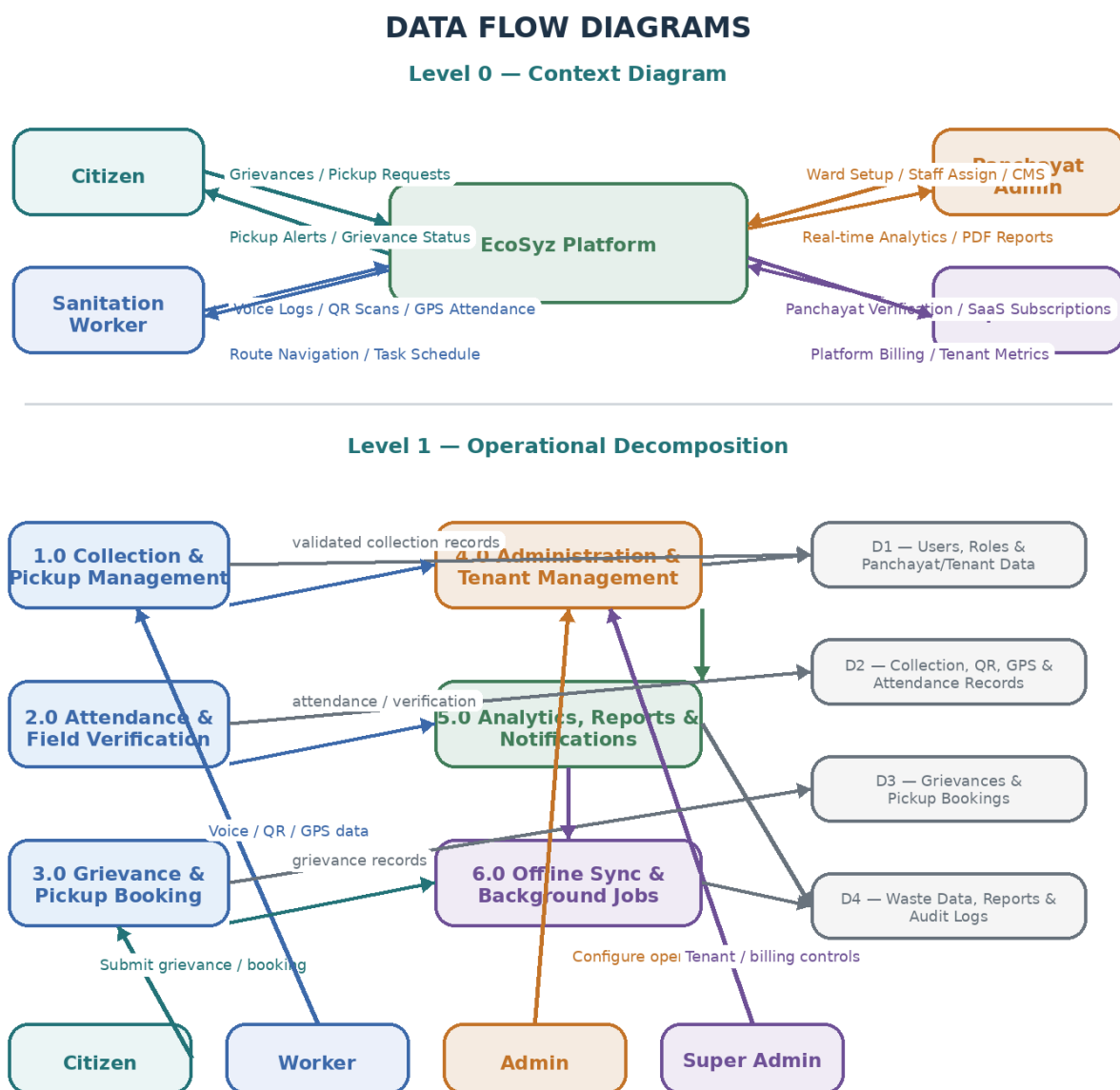


Figure 3.9. Data Flow Diagrams — Level 0 Context Diagram and Level 1 Operational Decomposition

3.10 VOICE COLLECTION IMPLEMENTATION FLOW

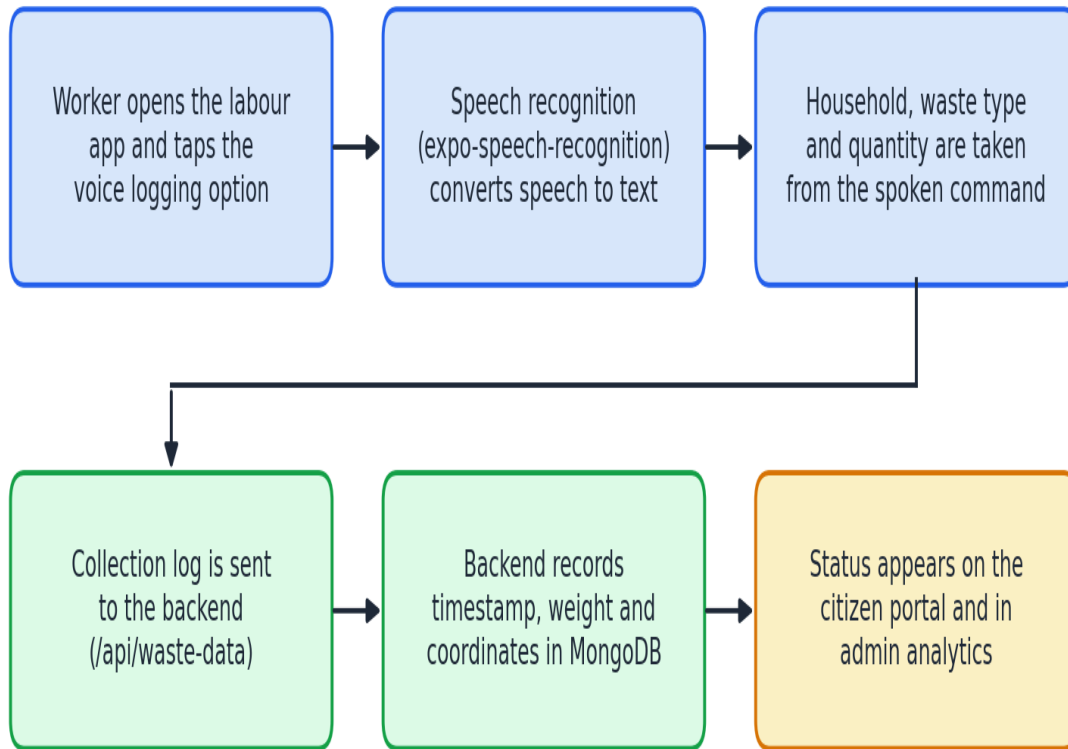


Figure 3.10. Voice Collection Processing Flow

3.11 BUILD AND CONNECTIVITY FLOW

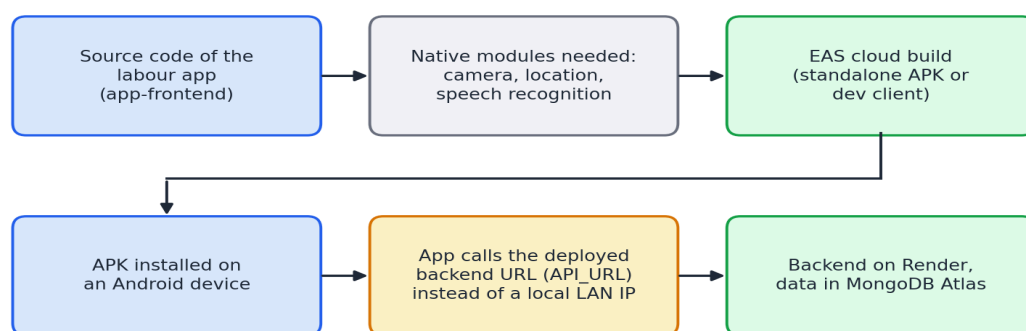


Figure 3.11. Mobile Build and Connectivity Flow

3.12 CORE IMPLEMENTATION

3.12.1 VOICE TOKENIZATION AND NLP

The voice collection workflow receives a speech transcript and extracts the household or bin number, waste category, and weight. The extracted values are converted into a structured collection payload and displayed to the worker before submission. This reduces manual typing while preserving a confirmation step.

Table 3.1. Voice collection processing flow

Input	Processing	Output
Speech transcript	Speech recognition	Text transcript
Transcript	Pattern matching / NLP	House number, waste type, weight
Structured values	Validation and confirmation	Collection payload
Payload + GPS	Backend validation	Stored collection record

3.12.2 GPS GEOFENCING

The backend uses the Haversine distance calculation to compare the worker coordinates with the registered service location. If the distance is within the configured boundary, the operation is accepted; otherwise the system returns an out-of-bound response.

3.12.3 QR COLLECTION

Each registered dustbin can be associated with a QR identifier. The mobile scanner reads the identifier, validates it with the backend, checks the collection state, and records the collection when the request satisfies the required conditions.

3.12.4 OFFLINE SYNCHRONIZATION

When network connectivity is unavailable, collection information can be retained locally. The application synchronizes the pending records after connectivity is restored, reducing the risk of losing field data in areas with unstable mobile coverage.

CHAPTER 4: SYSTEM TESTING AND VALIDATION

Testing was performed across Unit, Integration, Field Hardware, and End-to-End (E2E) workflows. The following matrix consolidates the principal validation cases used for authentication, mobile NLP, geofencing, attendance automation, reporting, and API load testing.

4.1 COMPREHENSIVE TEST EXECUTION MATRIX

Table 4.1. Comprehensive test execution matrix

Test ID	Module	Test Scenario / Objective	Test Steps & Input Data	Expected Result	Actual Result	Status
TC-AUTH-01	Auth Engine	Verify OTP generation and timeout.	Request OTP for registered 9222222222.	6-digit OTP created with 5-minute expiry in DB.	OTP generated and logged to console/SMS in 120ms.	Passed
TC-VOICE-01	Mobile NLP	Recognize compound voice collection command.	Spoken: "Hey Eco bin 7907 collected 3.5 kg".	Tokenizer extracts B-7907, 3.5kg, collected.	Accurately parsed with isValid: true in 85ms.	Passed
TC-VOICE-02	Mobile NLP	Handle background street noise & slang.	Spoken: "dabba cleared weight four kilos".	Phonetics normalize "dabba/ben" to the bin reference and weight.	Correctly normalized and matched B-102, 4kg.	Passed
TC-GEO-01	Geo Engine	Reject QR scan when worker is >20m away.	Worker at coordinates 85m away from Bin B-101.	System logs proximity warning and records distance.	Haversine computed 84.8m; warning logged.	Passed
TC-CRON-01	Attendance	Automated midnight roster generation.	Advance server clock to 00:00 AM.	All active staff receive default absent records.	14 records created across 2 panchayats seamlessly.	Passed
TC-PDF-01	Admin Web	Generate 30-day compliance report.	Select date range and click "Export PDF".	Multi-page PDF with footer, pagination and totals.	Generated crisp 3-page PDF in 340ms via jsPDF.	Passed
TC-LOAD-01	API Server	Concurrent scan stress test.	250 simulated workers scanning bins simultaneously.	0 dropped requests; p99 latency ≤250ms.	p99 latency 142ms; 0 failed transactions.	Passed

4.2 SECURITY ARCHITECTURE AND HARDENING MEASURES

- Role-Based Access Control (RBAC): JWT middleware verifies user roles before allowing access to administrative or protected endpoints.
- Multi-Tenant Data Isolation: database reads and writes are scoped by the authenticated Panchayat/tenant context to reduce the risk of cross-tenant data exposure.
- NoSQL Injection Prevention: strict schema validation and query sanitization are used for database inputs.
- File Storage Validation: uploaded files are checked for supported MIME types and size limits before storage.

CHAPTER 5: FUTURE ENHANCEMENTS AND CONCLUSION

5.1 FUTURE ENHANCEMENTS

1. IoT-enabled smart bins for fill-level monitoring and route planning.
2. Computer vision for waste segregation and contamination detection.
3. Citizen notifications through messaging platforms.
4. Broader integration with statewide waste-management monitoring.

5.2 CONCLUSION

EcoSyz provides a connected digital workflow for waste collection, attendance, field verification, citizen grievances, and administrative reporting. The implemented features demonstrate how web, mobile, GPS, QR, voice input, and cloud services can be combined into a practical waste-management reporting platform.

REFERENCES

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- Next.js Documentation, App Router and application development documentation.
- React Native Documentation, mobile application development reference.
- Expo Documentation, speech recognition, camera and geolocation references.
- MongoDB Atlas Documentation, cloud database and indexing reference.
- Node-Cron Documentation, task scheduling in Node.js applications.
- Tailwind CSS Documentation, utility-first interface framework reference.

APPENDIX I: USER MANUAL AND SCREEN DOCUMENTATION

This appendix documents the actual EcoSyz interface material available from the project report. The screenshots below are used directly from the supplied report so that the documentation represents the implemented interface rather than placeholder diagrams.

SCREEN DOCUMENTATION 1

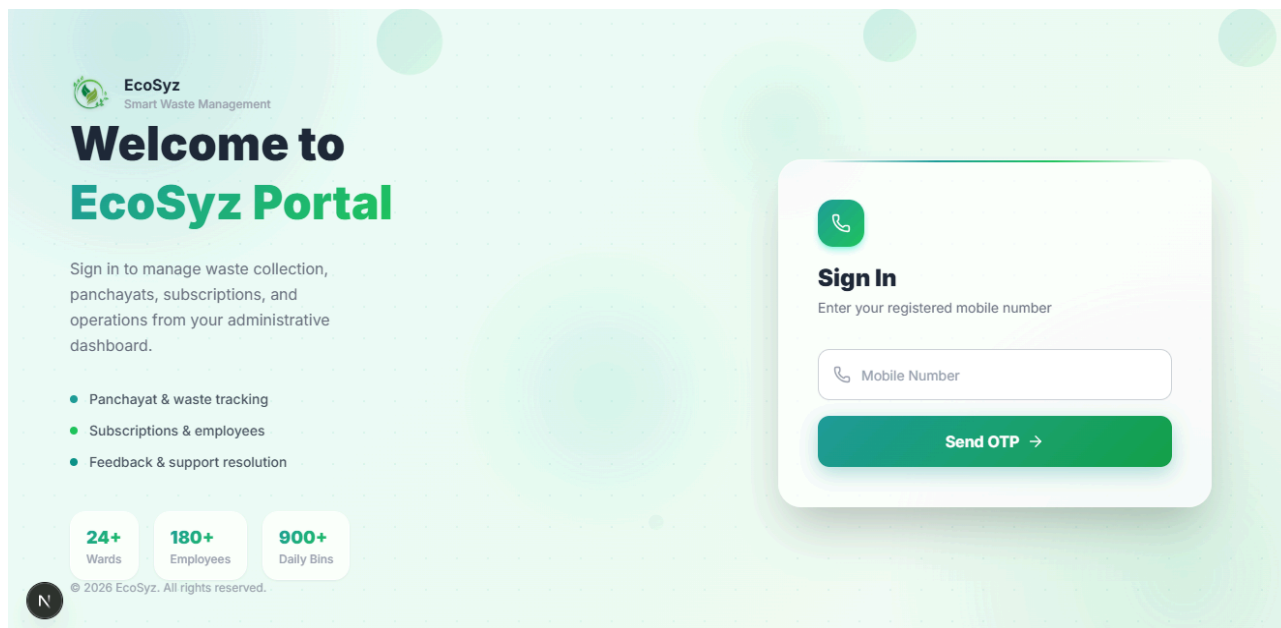


Figure A1.1. Unified login page for admin and super admin portal

SCREEN DOCUMENTATION 2

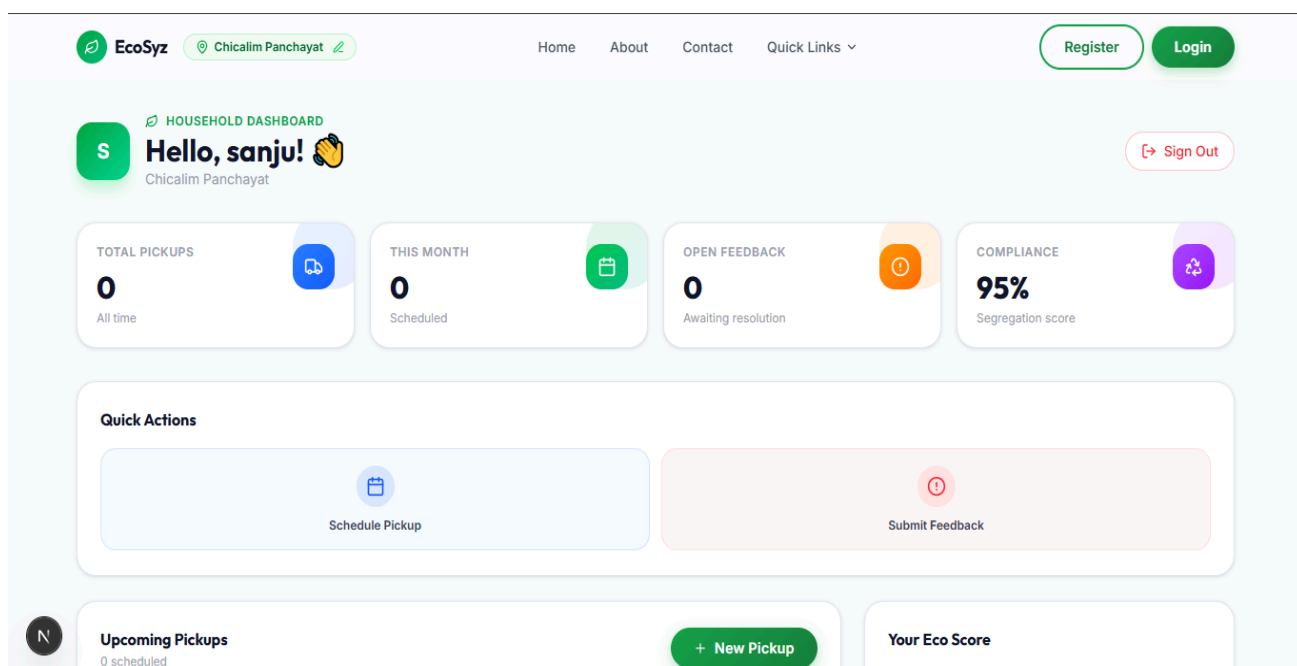


Figure A1.2. Citizen Dashboard

Citizen-facing dashboard showing service information and collection-related statistics.

SCREEN DOCUMENTATION 3

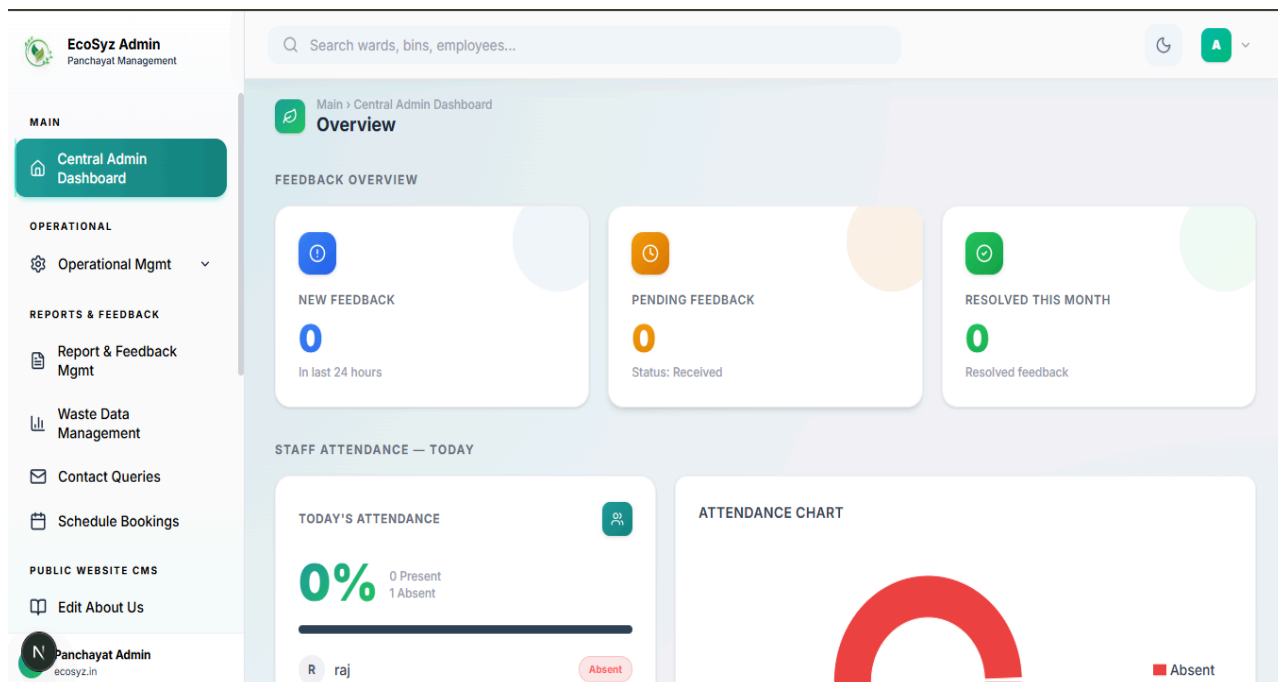


Figure A1.3. Panchayat Administrative Dashboard

Administrative dashboard used to monitor collection and operational statistics.

SCREEN DOCUMENTATION 4

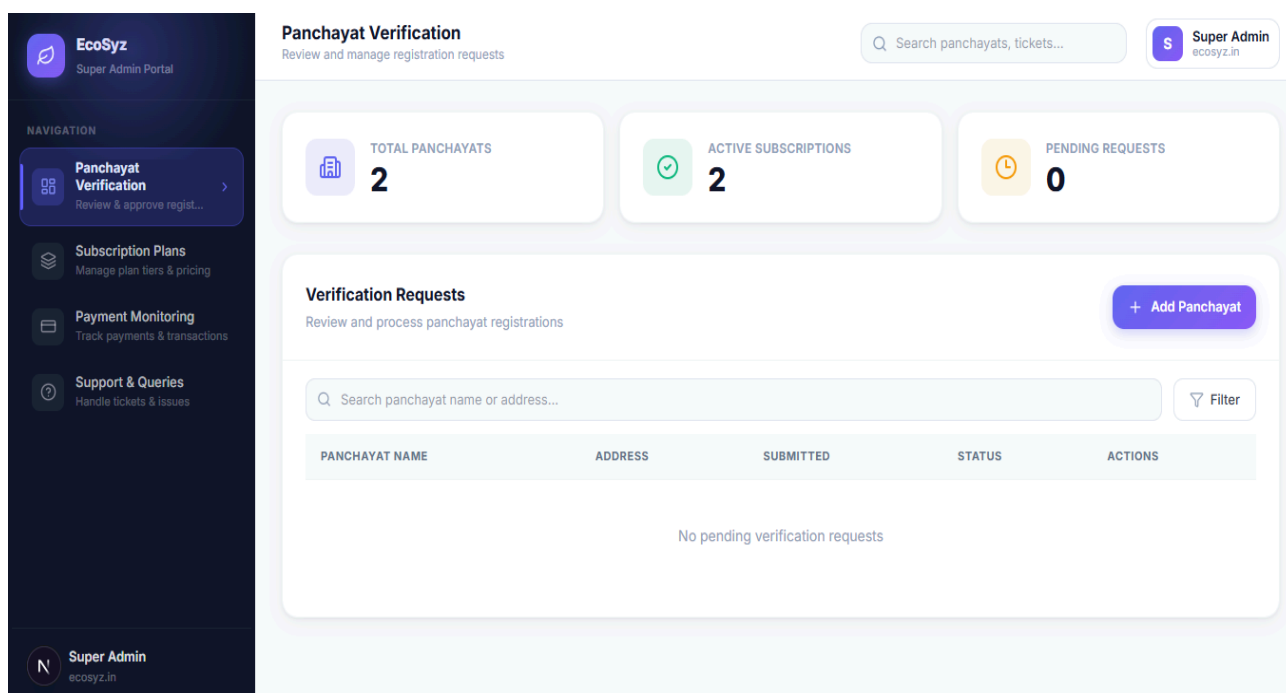
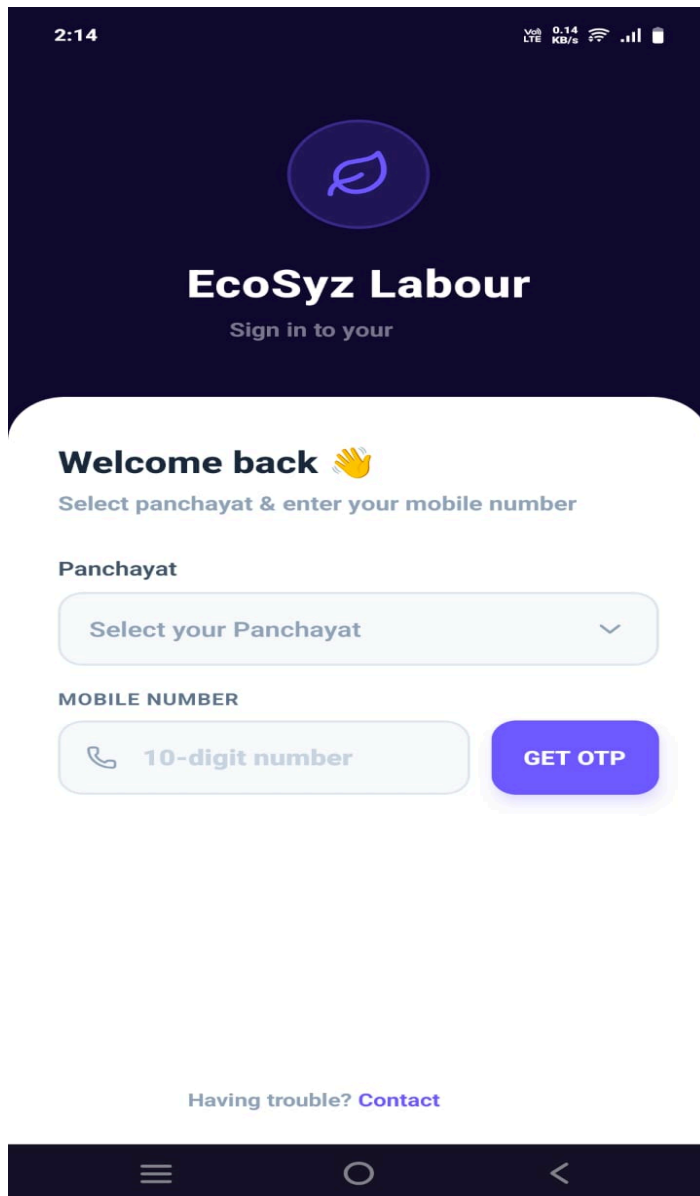


Figure A1.4. Panchayat Verification / Administrative Interface

Interface for administrative verification and management activity.

SCREEN DOCUMENTATION 5



The image shows a mobile application login screen for 'EcoSyz Labour'. At the top, the status bar displays the time 2:14, LTE signal, 0.14 KB/s data speed, and battery level. The app's logo, a stylized leaf inside a circle, is centered above the text 'EcoSyz Labour' and 'Sign in to your'. Below this, a 'Welcome back' message with a hand icon is followed by the instruction 'Select panchayat & enter your mobile number'. The 'Panchayat' section features a dropdown menu with the placeholder 'Select your Panchayat'. The 'MOBILE NUMBER' section has a text input field with a phone icon and the placeholder '10-digit number', and a blue 'GET OTP' button. At the bottom, a link 'Having trouble? Contact' is visible. The Android navigation bar is at the very bottom.

2:14

Vol 0.14 KB/s



EcoSyz Labour

Sign in to your

Welcome back 🖐️

Select panchayat & enter your mobile number

Panchayat

Select your Panchayat

MOBILE NUMBER

 10-digit number

GET OTP

Having trouble? [Contact](#)

Figure A1.5. Labour Application Login

Mobile login interface for field workers.

SCREEN DOCUMENTATION 6

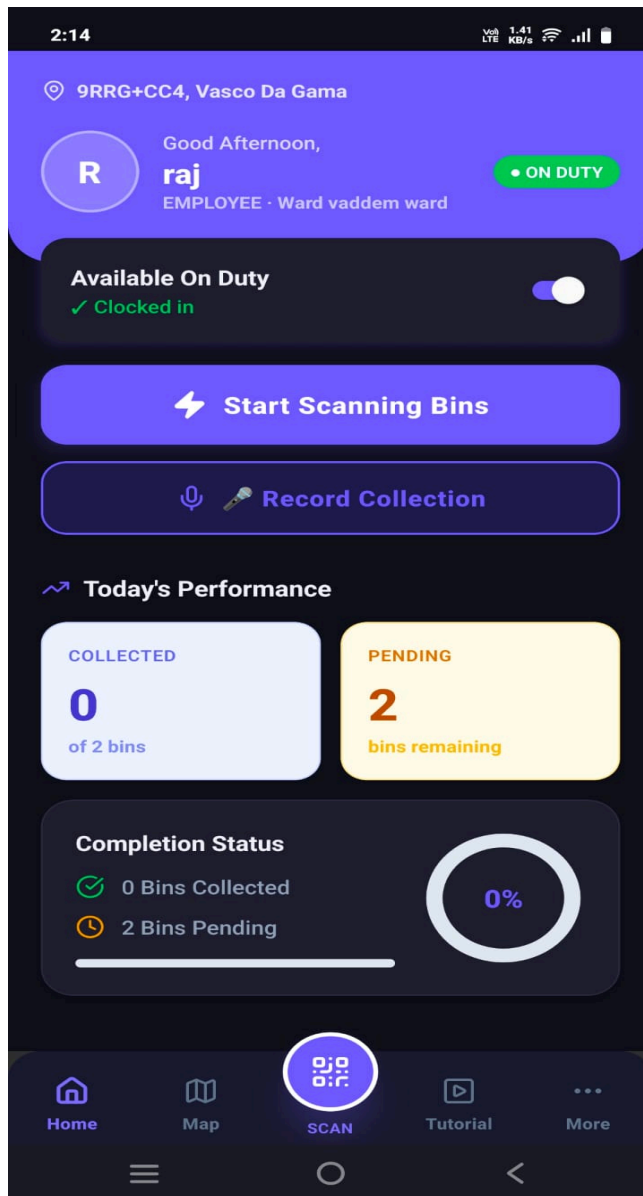


Figure A1.6. Labour Application Home

Worker home screen containing duty and collection-related controls.

SCREEN DOCUMENTATION 7

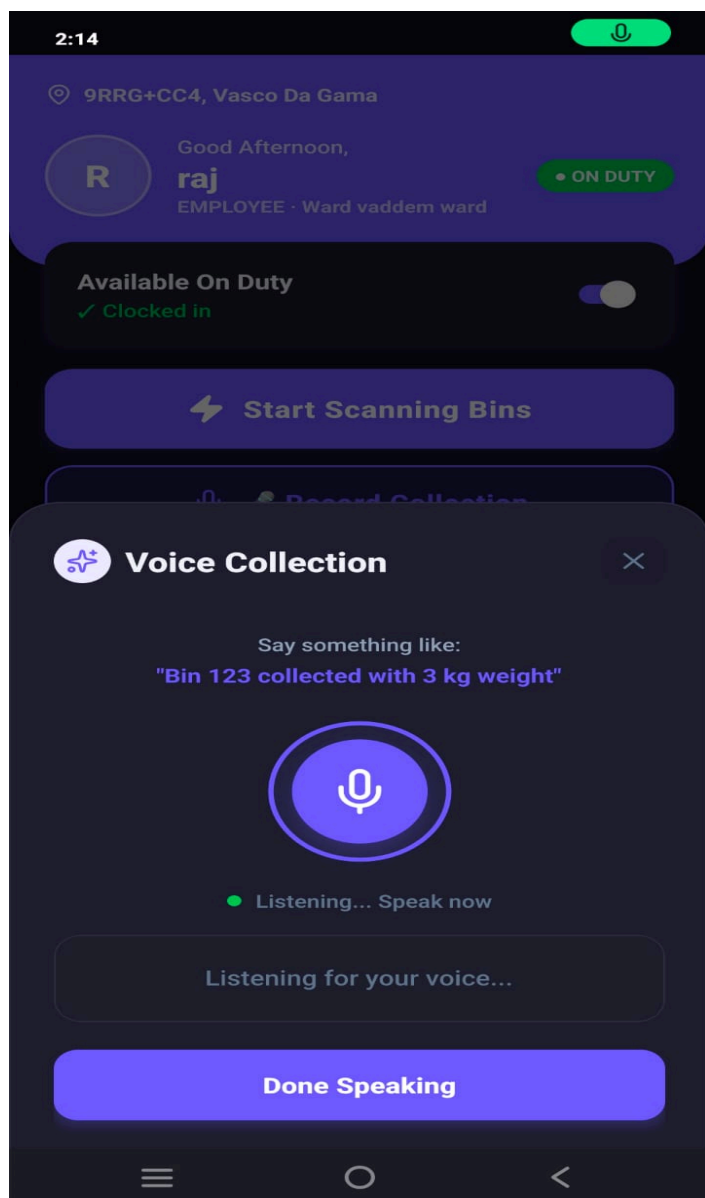


Figure A1.7. Voice Collection Modal

Voice-assisted collection screen used to enter waste information.

SCREEN DOCUMENTATION 8

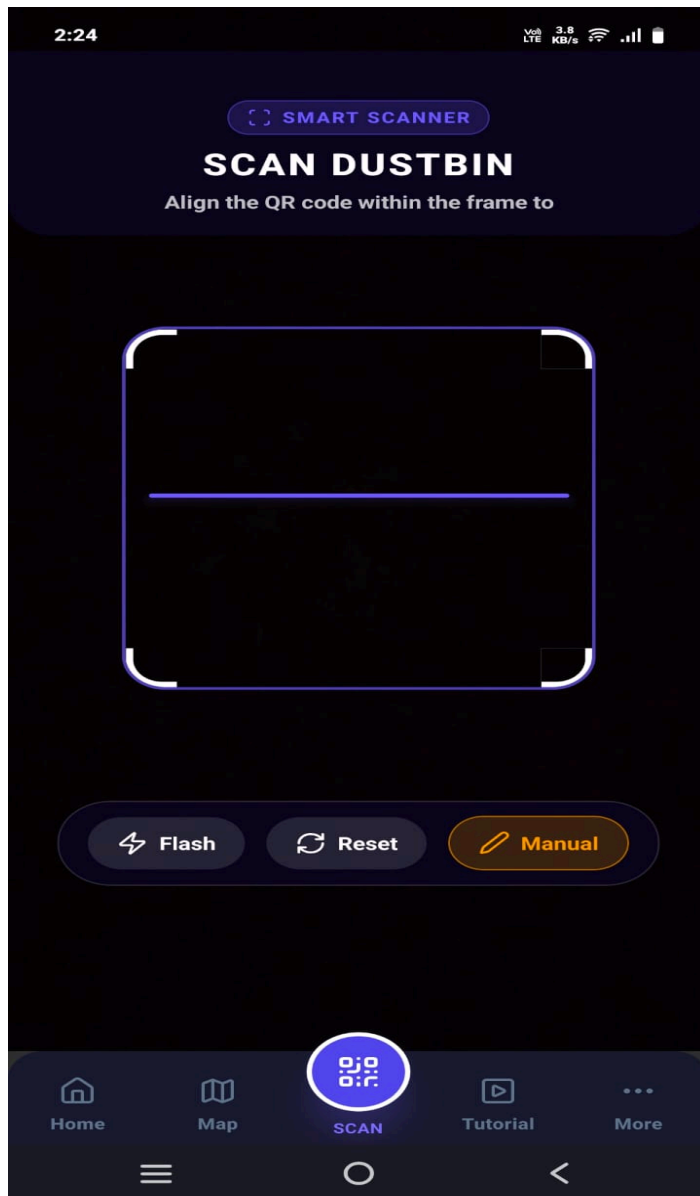


Figure A1.8. QR Scanner

QR scanning interface used for dustbin/collection verification.

SCREEN DOCUMENTATION 9

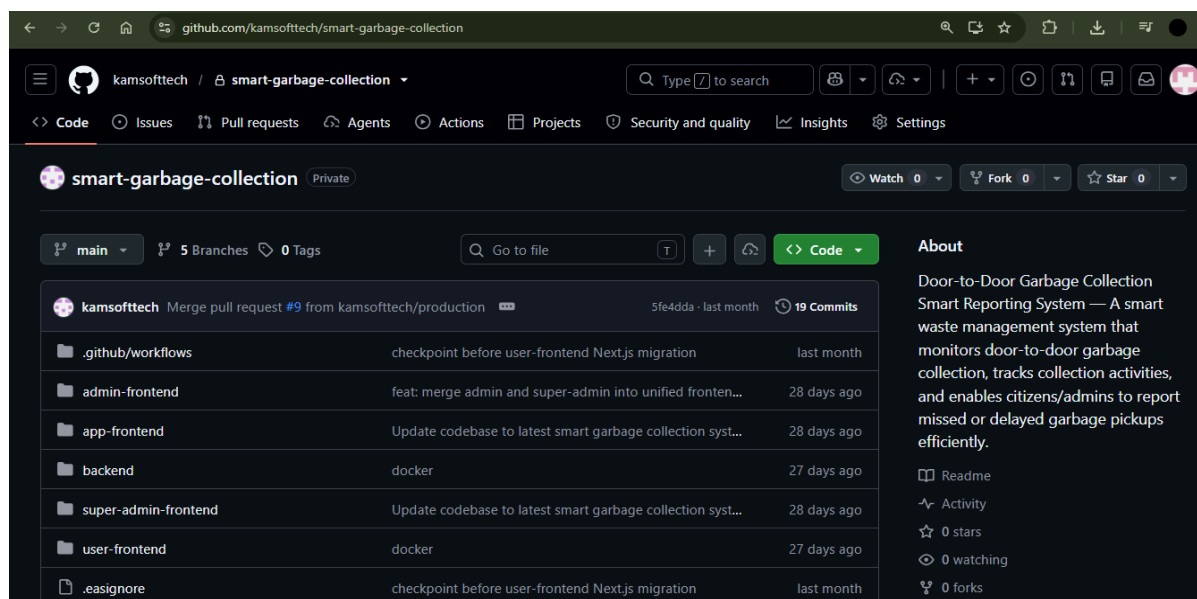


Figure A1.9. Project Repository

Project repository interface showing source-code organization and project files.

APPENDIX I – FUNCTIONAL SCREEN SUMMARY

Table A1.1. Functional screen summary

Module	Main Functions
Citizen Portal	Panchayat selection, home page, registration, OTP login, schedule viewing, complaint submission, guides and citizen dashboard.
Panchayat Admin	Dashboard, employee management, attendance, household/dustbin registry, routes, complaints, waste data, reports and CMS.
Super Admin	Panchayat verification, subscriptions, payments, profile management and support administration.
Labour App	OTP login, on-duty attendance, route map, voice collection, QR scanning, offline synchronization and support.

APPENDIX II: PROJECT TIMELINE

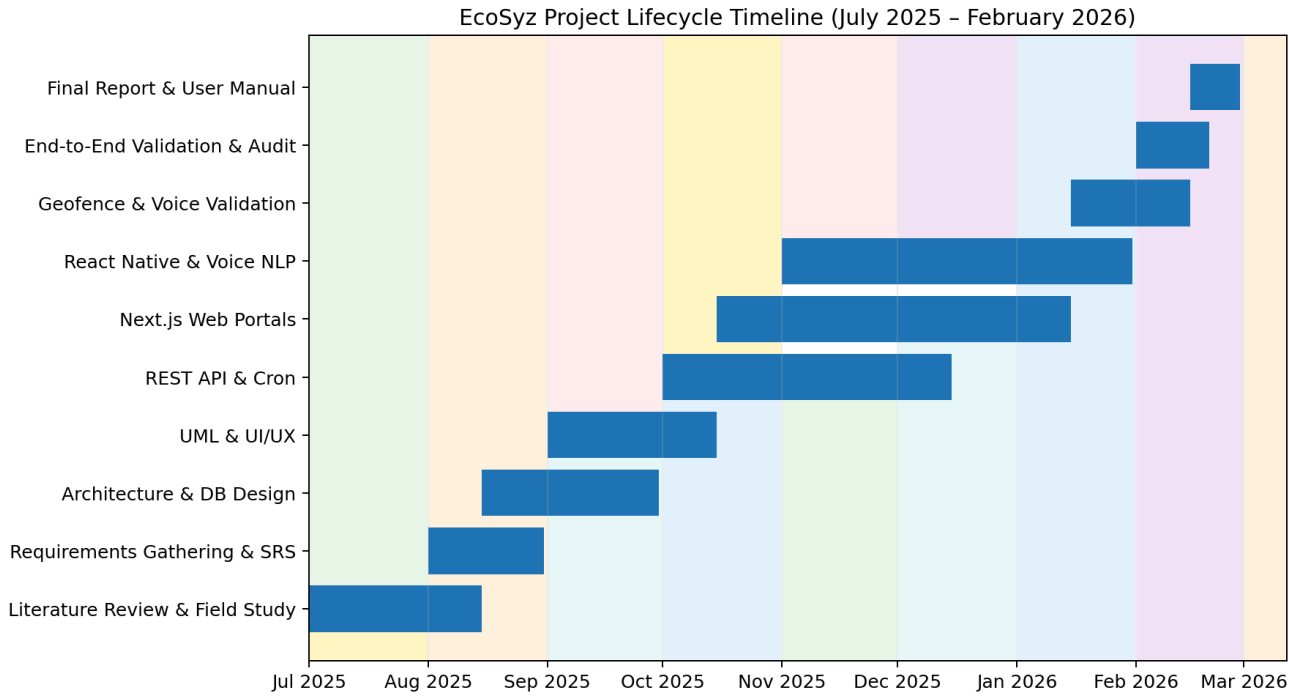


Figure A2.1. Project Gantt Chart Timeline

The timeline represents the major stages of analysis, design, development, testing, documentation, and demonstration of the EcoSyz project.